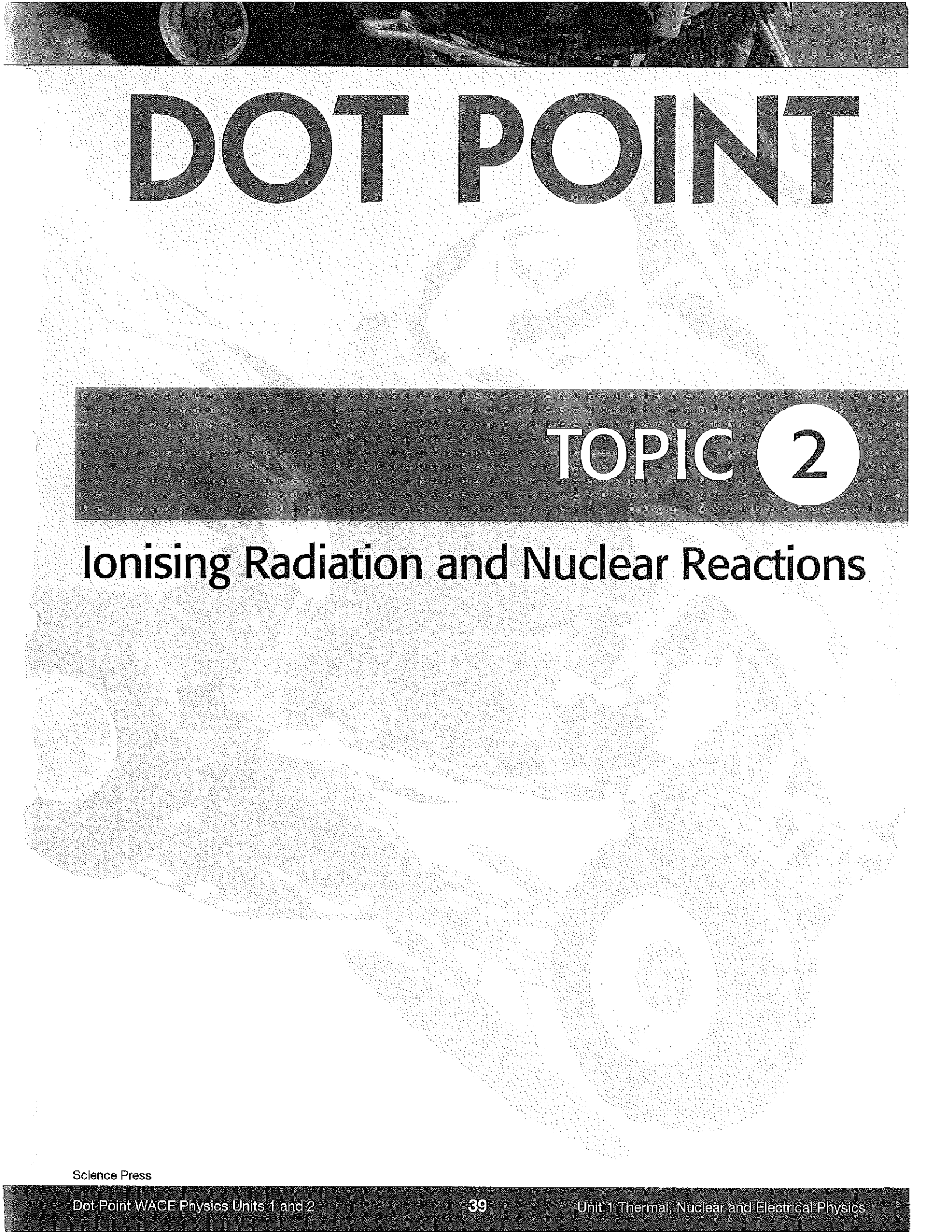




DOT POINT

TOPIC 2

Ionising Radiation and Nuclear Reactions

2.1 The nuclear model of the atom, models and theories.

2.1.1 Models of the atom 1.

2.1.1.1

- (a) Over the years several atomic models have been proposed. On what basis have these models been developed?

- (b) What is the reason that earlier models are replaced by later models?

- (c) What is the main feature of the nuclear models of the atom?

2.1.1.2

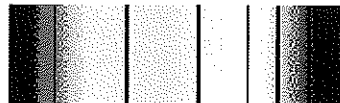
- (a) Who proposed the first nuclear model of the atom?
- (b) Describe the aspect of the experiment he did that led him to propose a nuclear model.

- (c) List four limitations of this nuclear model of the atom.

2.1.1.3

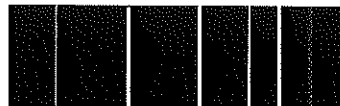
- (a) What was the main observation scientists were trying to explain in their development of successive nuclear models of the atom?

- (b) Although it is more difficult to distinguish which is which from a greyscale diagram, identify the three different types of spectrum represented by the three diagrams shown.



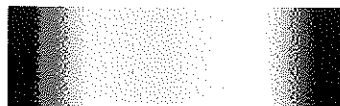
A

- (c) Briefly describe how each of these different types of spectra form.



B

A



C



B

.....

.....

C

.....

.....

- (d) Explain why the black lines in spectrum A above are in exactly the same positions as the whitish lines in spectrum B above.
-
-
-

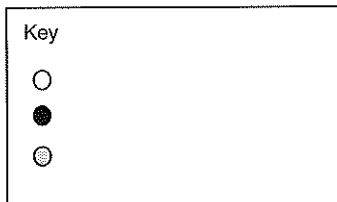
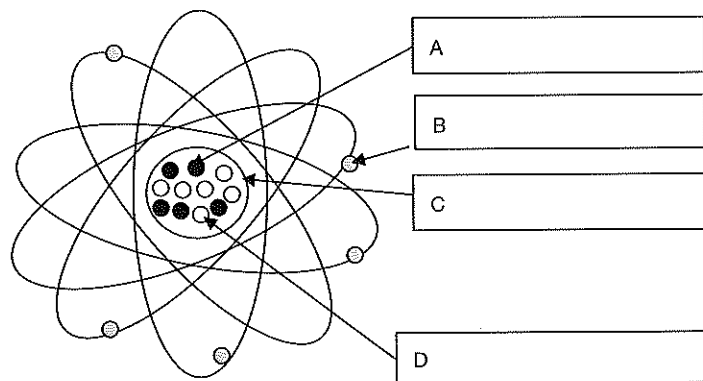
2.1.2 Models of the atom 2.

2.1.2.1

- (a) Define atomic number.
-
- (b) Define mass number.
-
- (c) Define atomic mass.
-

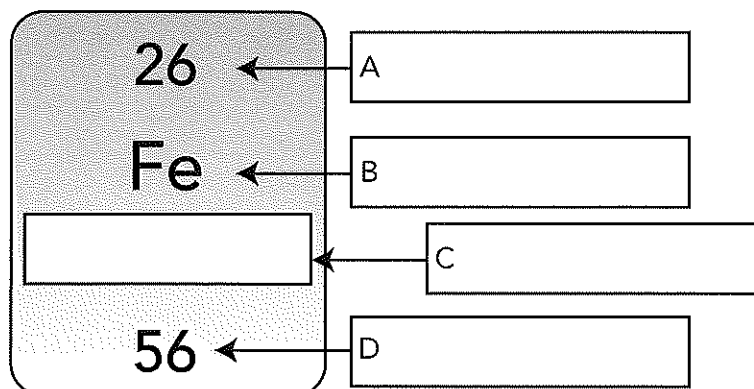
2.1.2.2

- (a) Consider the following diagram.
- (b) What is the atomic number of the element this atom represents?
-
- (c) What is the mass number of the element this atom represents?
-
- (d) Use the periodic table of the elements at the back of the book to identify the element this atom represents.
-
-
- (e) How does this atom differ from the atom of the next element in the periodic table? Justify your answer.
-
-



2.1.2.3

- (a) Identify the missing information in the following diagram.



- (b) What is the atomic number of this element?
- (c) What is the mass number of this element?

2.1.2.4 The property which distinguishes one element from another is:

- (A) The number of protons in the nuclei of the atoms.
 (B) The number of neutrons in the nuclei of the atoms.
 (C) The number of electrons in orbit around the nuclei.
 (D) The mass of their atoms.

2.1.2.5 An atom of sodium (Na) has 11 protons and 12 neutrons. Which formula is correct?

- (A) $^{11}_{12}\text{Na}$ (B) $^{11}_{23}\text{Na}$ (C) $^{12}_{11}\text{Na}$ (D) $^{23}_{11}\text{Na}$

2.1.2.6 The table shows information about four elements.

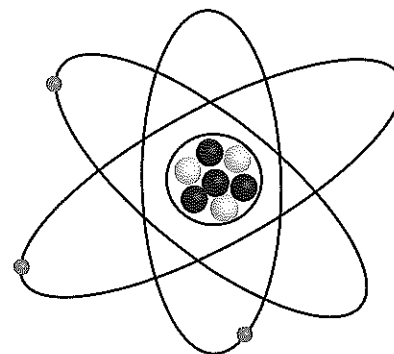
How many protons in each atom of Y?

- (A) 5
 (B) 27
 (C) 32
 (D) 59

	Number of protons	Number of neutrons	Number of electrons
W	9	10	9
X	16	16	16
Y	?	32	27
Z	38	50	38

2.1.2.7 Which statement about this atom is correct?

- (A) It contains 3 protons, 3 electrons and 4 neutrons.
 (B) It contains 4 protons, 3 electrons and 3 neutrons.
 (C) It contains 3 protons, 4 electrons and 3 neutrons.
 (D) It contains 3 protons, 4 electrons and 4 neutrons.



2.1.2.8 Which choice is correct?

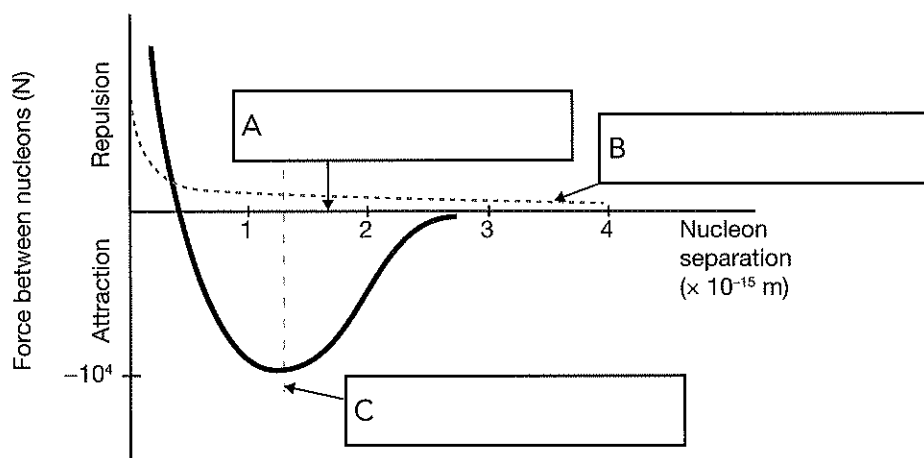
	Nucleon	Properties
(A)	Neutron	Positive, mass of one amu
(B)	Proton	Positive, mass of one gram
(C)	Electron	Negative, mass of one amu
(D)	Proton	Positive, mass of one amu

2.2 Nuclear stability.

2.2.1 The strong nuclear force.

2.2.1.1

- (a) What is the strong nuclear force?
-
-
- (b) Label the following graph.



- (c) List the four main properties of the strong nuclear force.

2.2.1.2 Explain why large nuclei are less stable than small nuclei.

2.3 Nuclides and decay.

2.3.1 Isotopes.

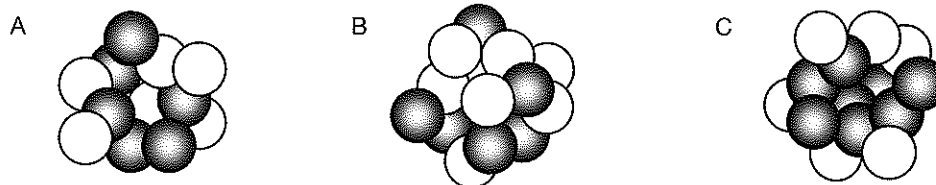
2.3.1.1

- (a) What are isotopes?
- (b) Comment on the atomic numbers of the different isotopes of a particular element. Explain your answer.
- (c) Comment on the mass numbers of the different isotopes of a particular element. Explain your answer.
- (d) Explain how isotopes affect the atomic mass of an element compared to its mass number.

2.3.1.2 Complete the table to show the structure of the isotopes of lithium, element number 3.

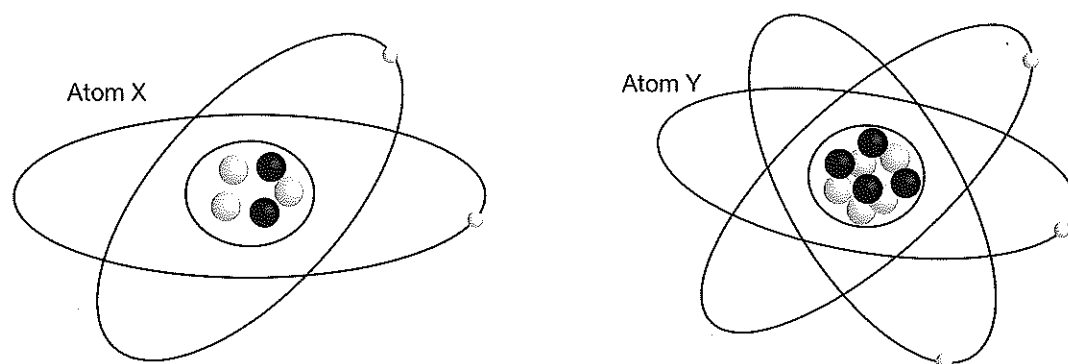
Isotope	Number of protons	Number of neutrons	Number of electrons	Atomic number	Mass number
Lithium-6					
Lithium-7					
Lithium-8					

2.3.1.3 The diagrams shows three isotopes of carbon, atomic number 6. Use the information in the diagrams to complete the table below.



Isotope	Number of protons	Number of neutrons	Number of electrons	Atomic number	Mass number
A					
B					
C					

2.3.1.4 A student drew diagrams to represent two different atoms.



- (a) What evidence is in the diagrams to indicate that these are atoms of *different* elements and not isotopes of the same element?

.....

.....

- (b) Assuming the nuclei have been drawn correctly, indicate an error in the student's diagrams and state how to correct it.

.....

.....

2.3.2 Nuclear decay.

- 2.3.2.1** Describe the following types of nuclear decay, indicating what happens to the nucleus as a result of each decay process.

- (a) Alpha decay.

.....

- (b) Beta decay.

.....

- (c) Neutron decay.

.....

- (d) Positron decay.

.....

- (e) Proton emission.

.....

- (f) Outline the situation with gamma emission.

.....

.....

2.3.2.2 On the following periodic table grid, indicate what happens to the atomic number (Z) and mass number (A) of an atom when it undergoes each type of nuclear decay (alpha, beta, positron, proton and neutron).

Identify the position of the new nucleus by drawing an arrow from the parent atom to the grid square which holds the new nucleus.

Identify each nuclear decay by labelling each arrow.

		Parent atom		

Atomic number

2.3.2.3

- (a) Using the format of the equation shown for alpha decay in the table, write a general equation for each of the other four types of nuclear decay.

Decay reaction	General equation for reaction
Alpha	${}^A_ZX \rightarrow {}^{A-4}_{Z-2}X + {}^4_2\text{He}$
Beta	
Positron	
Proton	
Neutron	

- (b) Explain why gamma decay has not been included in the table above.

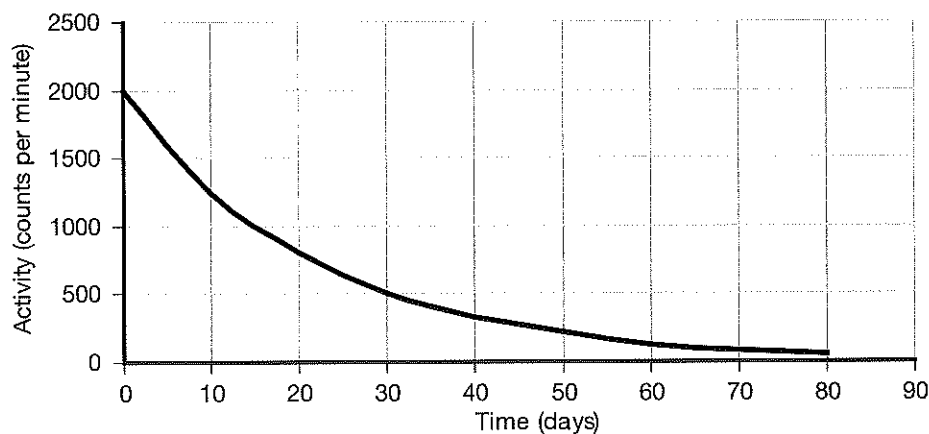
.....

.....

.....

- 2.3.2.4** Because energy is released during a radioactive emission, what change occurs in the nucleus?
 (A) The nucleus will decrease in mass. (B) The nucleus will become more stable.
 (C) The nucleus will increase in mass. (D) The nucleus will become unstable.
- 2.3.2.5** Which particle always accompanies positron emission?
 (A) Antineutrino. (B) Beta particle. (C) Gamma ray. (D) Neutrino.
- 2.3.2.6** Which of the following best describes a nuclear transmutation?
 (A) Elements emit radioactive particles to change into other elements.
 (B) Elements react with each other to produce new elements.
 (C) Reactions which involve the release of enormous amounts of energy.
 (D) Reactions in which elements combine to produce unknown elements.
- 2.3.2.7** Element Y undergoes two beta decays. Which statement about the product is correct?
 (A) Atomic number down 2, mass number up 2.
 (B) Atomic number up 2, mass number stays the same.
 (C) Atomic number up 2, mass number down 2.
 (D) Atomic number down 2, mass number stays the same.
- 2.3.2.8** Element X emits an alpha particle to form element Y. Which statement is true about X and an atom of Y?
 (A) Atoms of X have 4 more protons than atoms of Y.
 (B) Atoms of X have 2 less neutrons than atoms of Y.
 (C) Atoms of X have 2 more neutrons than atoms of Y.
 (D) Atoms of X have 2 less protons than atoms of Y.
- 2.3.2.9** Where does the energy released by a radioactive substance come from?
 (A) The breaking of chemical bonds between nucleons.
 (B) The breaking of chemical bonds within nucleons.
 (C) The breaking of bonds within the nucleus.
 (D) The mass defect of the nuclear emissions.

Use the following half-life graph to answer the next TWO questions.



- 2.3.2.10** Which is the best estimate for the half-life of the radioisotope from which this graph was drawn?
 (A) 10 days
 (B) 15 days
 (C) 20 days
 (D) 25 days
- 2.3.2.11** What would be the activity of the isotope after 25 days?
 (A) About 600 counts per minute.
 (B) About 625 counts per minute.
 (C) About 650 counts per minute.
 (D) About 675 counts per minute.

2.3.3 Stability of nuclides.

2.3.3.1 Use the concept of entropy to explain why some nuclei are unstable whereas others are stable.

.....

.....

.....

2.3.3.2 Use the idea of the distance between nucleons and the strong nuclear force to explain the relative stability of atoms.

.....

.....

.....

2.3.3.3 Explain why all isotopes of elements above lead, atomic number 82, tend to be unstable.

.....

.....

.....

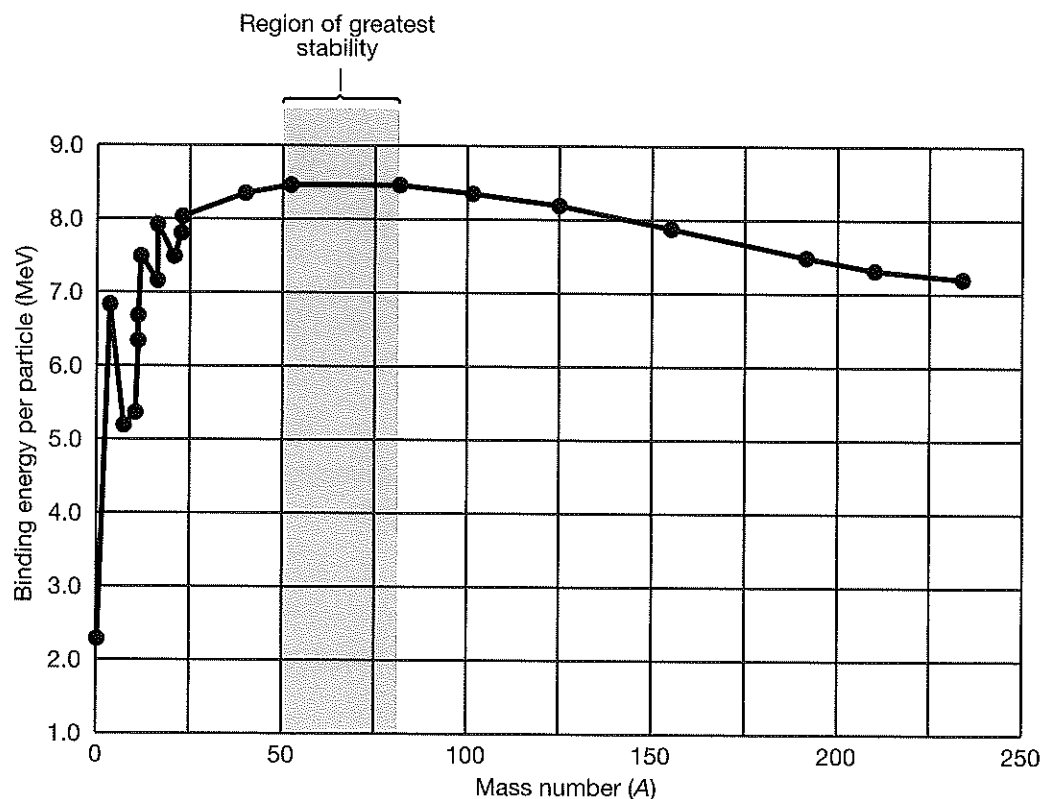
2.3.3.4 Scientists have observed that atoms with even numbers of both protons and neutrons tend to be the most stable while those with odd numbers of both are the least stable. What is the explanation scientists give for this?

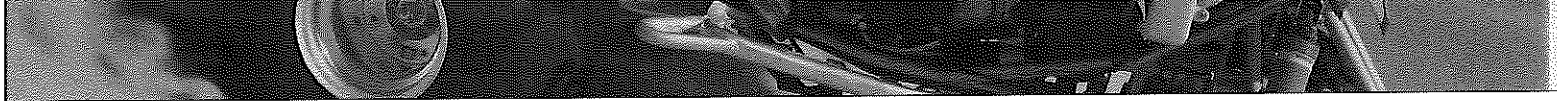
.....

.....

.....

2.3.3.5 Consider the following graph which relates the stability of isotopes to their mass number and binding energy.



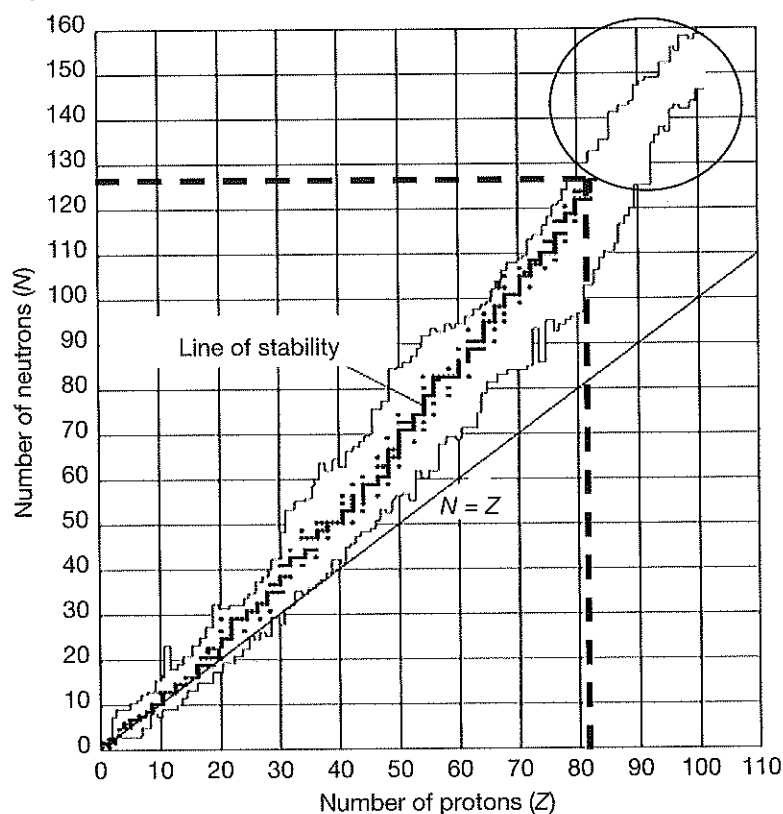


- (a) What is meant by binding energy?
-
- (b) What is this graph telling us in terms of stability and binding energy?
-
- (c) Explain the theory behind the region of greatest stability atoms.
-
- (d) Does this information tell us that atoms with mass numbers less than 50 or greater than 82 are unstable? Explain your answer.
-

2.3.3.6 The diagram is a typical 'band of stability' diagram.

Use it to answer this question.

- (a) What is the 'line of stability' shown on the diagram?
-
-
- (b) Atoms of isotopes above the line of stability tend to be unstable by beta decay. Explain, in terms of proton/neutron ratios why this is so.
-
-
-
-
- (c) Atoms of isotopes below the line of stability tend to be unstable by positron decay. Explain, in terms of proton/neutron ratios why this is so.
-
-



- (d) Atoms of isotopes beyond the line of stability (circled area) tend to be unstable by alpha decay. Explain, in terms of proton/neutron ratios why this is so.
-
-
- (e) What must we keep in mind when using this data to predict the decay of particular isotopes?
-

2.3.4 Properties of radioactive particles.

2.3.4.1 Complete the following table.

Radiation	Charge	Mass (amu)	Penetrating power	Ionising power	Path through electric field	Path through magnetic field	Stopped by	What is it?
Alpha								
Beta								
Gamma								

2.3.4.2

(a) What is meant by the term 'ionising radiation'?

.....

(b) In terms of the properties of the particles shown in the table above, explain the differing penetrating powers of these three radioactive emissions.

.....

(c) In terms of the properties of the particles shown in the table above, explain the differing ionising powers of these three radioactive emissions.

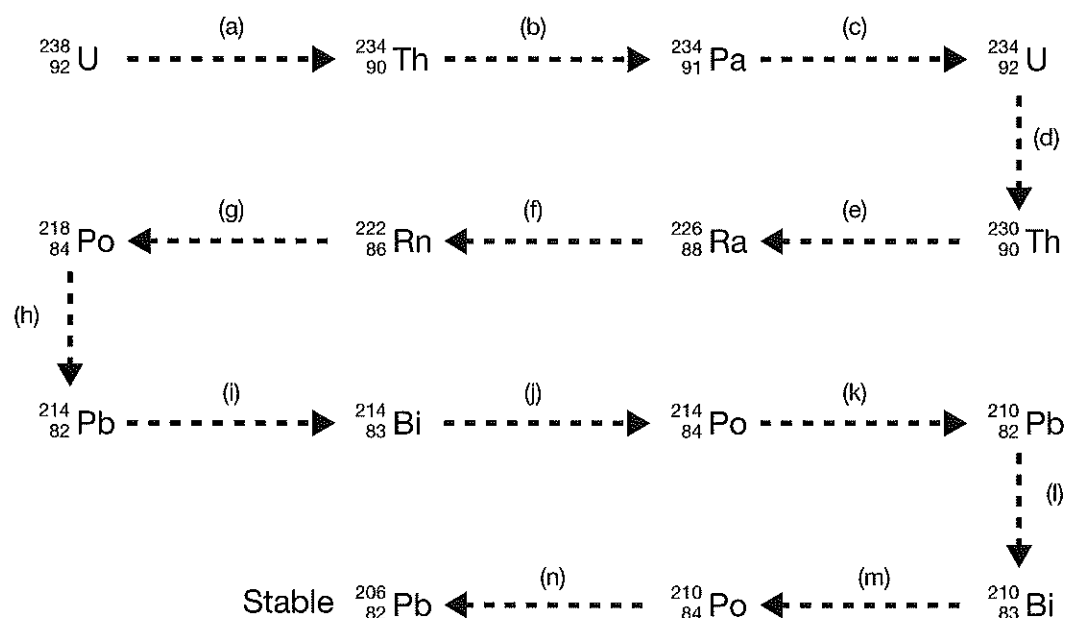
.....

2.3.4.3 The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise information about the three most common radioactive emissions.

Completed sentences		Second halves (not in order)
(a)	In alpha particle decay, a positively charged	centimetres in air.
(b)	This alpha particle consists of two	at the speed of light.
(c)	Alpha particles are	human skin.
(d)	Alpha particles can only travel a few	metres of air.
(e)	Alpha particles can be stopped	protons and two neutrons.
(f)	Alpha particles can be stopped by	a very high energy state.
(g)	In beta decay, an electron	millimetres of human skin.
(h)	Beta particles can penetrate several	several metres of water.
(i)	Beta particles can penetrate several	electromagnetic radiation.
(j)	Most beta particles will be stopped by a few centimetres of a light material	particle is emitted from the nucleus.
(k)	Gamma decay occurs because the nucleus of an atom is in	lead, steel or concrete.
(l)	The nucleus 'falls down' to a lower energy state, emitting a photon,	by a sheet of paper.
(m)	Gamma particles or gamma rays are actually	like aluminium or plastic.
(n)	Gamma particles travel	helium nuclei.
(o)	Gamma rays can be stopped by	more commonly known as a gamma ray.
(p)	Gamma rays can be stopped by	is emitted from the nucleus.

2.3.5 Writing equations for nuclear decay reactions.

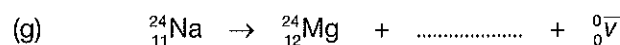
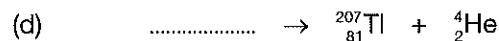
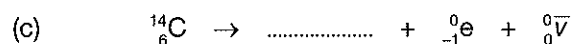
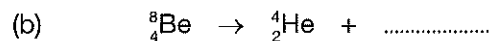
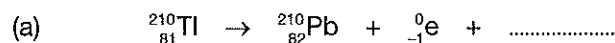
2.3.5.1 The flow diagram shows the sequence of nuclear decay processes that occur in the uranium-238 chain. The radiation emitted during each decay is not shown.



Write nuclear equations for each of the decay reactions in the uranium-238 chain.

(a)		(h)	
(b)		(i)	
(c)		(j)	
(d)		(k)	
(e)		(l)	
(f)		(m)	
(g)		(n)	

2.3.5.2 Some incomplete nuclear decay equations are shown. Identify the missing parts.



2.3.5.3 Atom P undergoes alpha decay to form atom Q. Compared to P, an atom of Q will have:

- (A) 2 protons more. (B) 2 protons less, same neutrons.
(C) 2 protons and 2 neutrons more. (D) 2 protons and 2 neutrons less.

2.3.5.4 Consider ${}_Z^AX \rightarrow {}_{Z+1}^AX + {}_{-1}^0\text{e} + Z$.
Z is best described as:

- (A) An antineutrino. (B) An alpha particle.
(C) A beta particle. (D) A neutrino.

2.3.5.5 Which choice correctly gives the nuclear equation for the reaction between element X and alpha particles to produce a neutron?

- (A) ${}_A^ZX + {}_2^4\text{He} \rightarrow {}_{A-2}^{Z+4}\text{X} + \gamma$
(B) ${}_A^ZX + {}_2^4\text{He} \rightarrow {}_{A-2}^{Z-3}\text{X} + {}_0^1\text{n}$
(C) ${}_A^ZX + {}_2^4\text{He} \rightarrow {}_{A+2}^{Z-3}\text{X} + {}_0^1\text{n}$
(D) ${}_A^ZX + {}_2^4\text{He} \rightarrow {}_{A-2}^{Z+3}\text{X} + {}_0^1\text{n}$

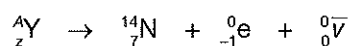
2.3.5.6 If an atom of bismuth emits an alpha particle, what element will be formed?

- (A) Francium. (B) Gold.
(C) Thallium. (D) Lead.

2.3.5.7 If an atom of polonium emits a beta particle, what element will be formed?

- (A) Astatine. (B) Bismuth.
(C) Mercury. (D) Radon.

2.3.5.8 What is the correct identification for element Y in this reaction.



- (A) ${}_{10}^{14}\text{N}$ (B) ${}_{10}^{14}\text{C}$ (C) ${}_{10}^{15}\text{P}$ (D) ${}_{10}^{14}\text{P}$

2.3.5.9 If iridium-77 undergoes two successive beta decays, what element is formed?

- (A) Gold. (B) Mercury.
(C) Osmium. (D) Rhenium.

2.3.5.10 What always accompanies the emission of a beta particle during nuclear decay?

- (A) An alpha particle. (B) An antineutrino.
(C) A gamma ray. (D) A neutrino.

2.4 Nuclides and half-life.

2.4.1 Half-life of nuclides.

2.4.1.1 Define the term 'half-life' as it applies to radioisotopes.

2.4.1.2 A particular radioactive isotope has a half-life of 4 hours and a 5 g specimen of this isotope in the laboratory has a radioactive count of 200 s^{-1} at 9.00 o'clock when school starts.

- (a) What will be the radioactive count for this isotope at 1.00 o'clock?
- (b) At what time would the count be 25 s^{-1} ?
- (c) How many half-lives would have past if 1.25 g of the isotope remained in the sample?

2.4.1.3 Thorium-227 has a half-life of 17.82 days. The half-life of iodine-131 is 8.040 days. A nuclear researcher starts with a mass of each that has an activity of 200 counts per minute.

(a) What would be the activity of the Th-227 after 5 half-lives?

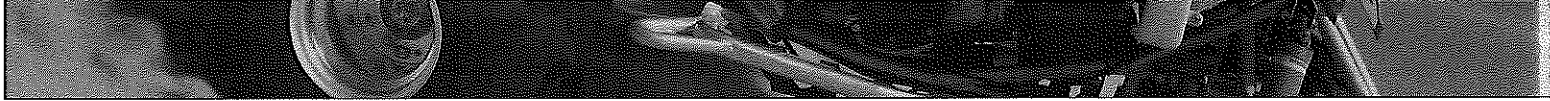
(b) How much time is represented by 5 half-lives for I-131?

(c) The researcher measured the activity of each sample after the same period of time. Which radioisotope would show the greatest activity? Justify your answer.

(d) What would be the activity of the thorium compared to the iodine after 50 days?

2.4.1.4 After 24.0 days, 2.00 mg of a 128.0 mg sample of an isotope remains. What is its half-life?

2.4.1.5 100.0 g of an isotope with a half-life of 36.0 hours is present at time zero. How long before only 5.00 g remains?



2.4.1.6 How much time will be required for a sample of H-3 to lose 75% of its radioactivity? The half-life is 12.26 years.

.....

.....

2.4.1.7 The half-life of iodine-131 is 8.040 days. What percentage remains after 40.2 days?

.....

.....

2.4.1.8 The half-life of thorium-227 is 18.72 days. How long for 62.5% of a sample to decay?

.....

.....

2.4.1.9 Rn-222 has a half-life of 3.82 days. How long before only one quarter of the sample remains?

.....

.....

2.4.1.10 You have 20.0 grams of ^{32}P that decays 5% daily. What is its half-life?

.....

.....

2.4.1.11 If you start with 5.32×10^9 atoms of Cs-137, how much time will pass before the amount remaining is 5.20×10^6 ? The half-life of Cs-137 is 30.17 years.

.....

.....

2.4.1.12 The half-life of the radioactive isotope phosphorus-32 is 14.3 days. How long until a sample loses 99% of its radioactivity?

.....

.....

2.4.1.13 A radioactive sample contained 3.25×10^{18} atoms of a nuclide that decayed at a rate of 3.4×10^{13} disintegrations in 26 minutes. (Assume that the rate of disintegrations will remain constant for this question – note that this is actually an incorrect assumption.)

(a) What percentage of the nuclide will have decayed after 159 days?

.....

.....

(b) What is the half-life of the nuclide?

.....

.....

2.4.2 Analysing a half-life experiment.

2.4.2.1 A nuclear scientist was investigating the properties of a radioisotope, X. Before starting the experiment she measured the background radiation and gained the results shown in the table.

- (a) Calculate the average background radiation in counts per minute (counts min⁻¹). Use your calculation to complete the table.

Time (s)	Number of counts
60	48
60	51
60	53
60	52
60	46
Average count	

- (b) Why is it necessary to measure this background radiation before starting the experiment?

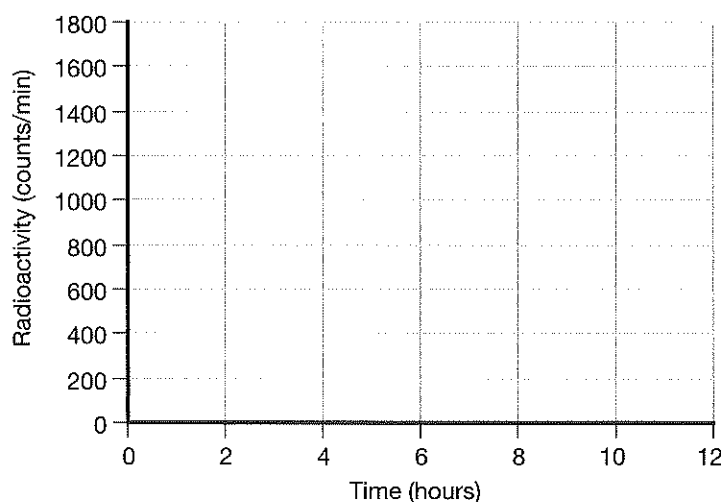
The scientist then placed a sample of X near the counter and measured the radiation each Monday for 7 weeks, counting the first week as week 0, the starting week. Her results are shown in the table.

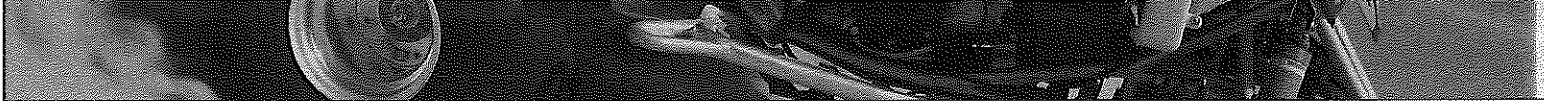
Time (hours)	Total radiation counts recorded (counts min ⁻¹)	Radiation due to X (counts min ⁻¹)
0	1730	
2	1110	
4	715	
6	465	
8	320	
10	215	
12	160	

- (c) Complete the table by writing in the values for the third column.
- (d) Plot the results to show the relationship between time and the radiation of X.
- (e) Use your graph to determine the number of days for the radiation count from X to halve, that is, the 'half-life' of X.

- (f) Predict the radiation in counts min⁻¹ of X after 20 hours.

- (g) Predict the radiation in counts min⁻¹ of X after 36 hours.





2.5 Advances in science understanding, ICT and technology.

2.5.1 Radioisotopes and radiometric dating.

2.5.1.1

- (a) What is radiometric dating?
- (b) What is radiometric dating used for?
- (c) Outline the principle behind radiometric dating.
- (d) Who invented the process of radiometric dating and when?
- (e) What was this scientist trying to do using radiometric dating?

2.5.1.2

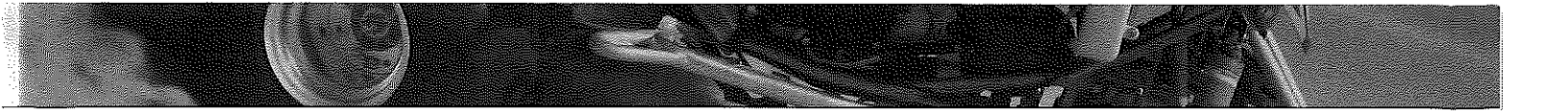
- (a) One of the most useful tools a geologist has is the geological time scale. What is this?
- (b) What role does radiometric dating play in using and developing the geological time scale?

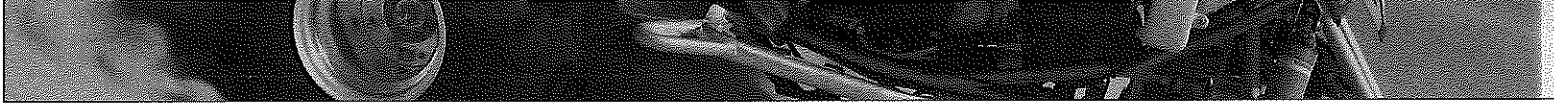
2.5.1.3

- (a) What radiometric dating method is used to date previously living organisms?
- (b) Outline the principle behind this method.

2.5.1.4

- (a) The half-life of C-14 is 5730 years. How much of this isotope would remain in a sample which was 63 030 years old?
- (b) Explain why this is the limit for dating using C-14.

- 
- 2.5.1.5** K-40 decays to Ar-40 with a half-life of 1.27×10^9 years. A sample of Moon rock has 78 argon-40 atoms for every 22 potassium-40 atoms. Assume the sample was 100% K-40 at the start. Find the age of the Moon rock.
-
- 2.5.1.6** What is the age of a rock in which the mass ratio of Ar-40 to K-40 is 3.8 : 1? K-40 decays to Ar-40 with a half-life of 1.27×10^9 years.
-
- 2.5.1.7** A sample of petrified wood has a C-14 decay rate of $6.00 \text{ counts min}^{-1}$. The decay rate of carbon-14 in fresh wood is $13.6 \text{ counts min}^{-1}$. If the half-life of C-14 is 5730 years, what is the age of the piece of wood?
-
- 2.5.1.8** The carbon-14 decay rate of a sample obtained from a young tree is 0.296 s^{-1} . Another sample prepared from an object recovered at an archaeological dig gives a decay rate of 0.109 s^{-1} . What is the age of the object?
-
- 2.5.1.9** The C-14 content of an ancient piece of wood was found to have three tenths of that in living trees (indicating 70% of the C-14 had decayed). How old is that piece of wood?
-
- 2.5.1.10** Carbon-14 is used to determine the age of ancient objects. If a sample today contains 0.060 mg of carbon-14, how much carbon-14 must have been present in the sample 11 430 years ago?
-
- 2.5.1.11** A living plant contains approximately the same isotopic abundance of C-14 as does atmospheric carbon dioxide. The observed rate of decay of C-14 from a living plant is $15.3 \text{ disintegrations min}^{-1}$. How many disintegrations min^{-1} will be measured from a 10 000-year-old sample? (The half-life of C-14 is 5730 years.)
-
- 2.5.1.12** All current plants have a C-14 count of $15.3 \text{ counts min}^{-1}$. How old is a wooden artefact if it has a count of $9.58 \text{ counts min}^{-1}$?
-
- 2.5.1.13** Using dendrochronology (using tree rings to determine age), tree materials dating back 10 000 years have been identified. Assuming you had a sample of such a tree in which the number of C-14 decay events was 15.3 decays per minute before decomposition, what would the decays per minute be in the present day?
-



2.6 Alpha, beta and gamma radiation.

2.6.1 Effects of radiation.

2.6.1.1 Define each of the four measures used when studying radioactivity giving their SI unit and a definition of each unit.

(a) Activity.
.....
.....

(b) Exposure.
.....
.....

(c) Absorbed dose.
.....
.....

(d) Dose equivalent.
.....
.....

2.6.1.2 Describe the factors which influence the biological effects of radiations listed below.

(a) Type of radiation involved.
.....
.....

(b) Size of dose received.
.....

(c) Rate the dose is received.
.....

(d) Part of the body exposed.
.....

(e) The age of the individual.
.....

(f) Biological differences.
.....

2.7 Einstein's mass-energy relationship.

2.7.1 Nuclear reactions.

2.7.1.1

(a) What is the binding energy of a nucleus?

(b) Compare the relative strength of the binding energy in a nucleus to the electrostatic force holding electrons in orbit (ball park answer will suffice).

(c) How does the binding energy per nucleon in a nucleus relate to the stability of the nucleus?

2.7.1.2 If the binding energy per nucleon of $^{62}_{28}\text{Ni}$ is 8.7948 MeV, what is its total binding energy:

(a) In MeV?

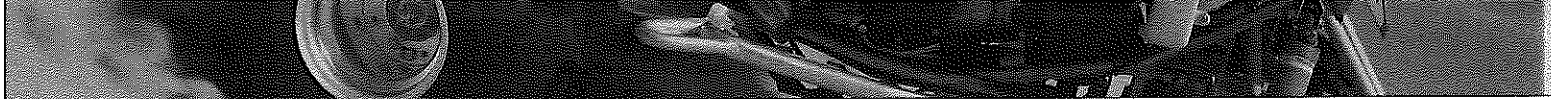
(b) In joules?

2.7.1.3 Consider nuclei of $^{31}_{15}\text{P}$ and $^{39}_{19}\text{K}$.

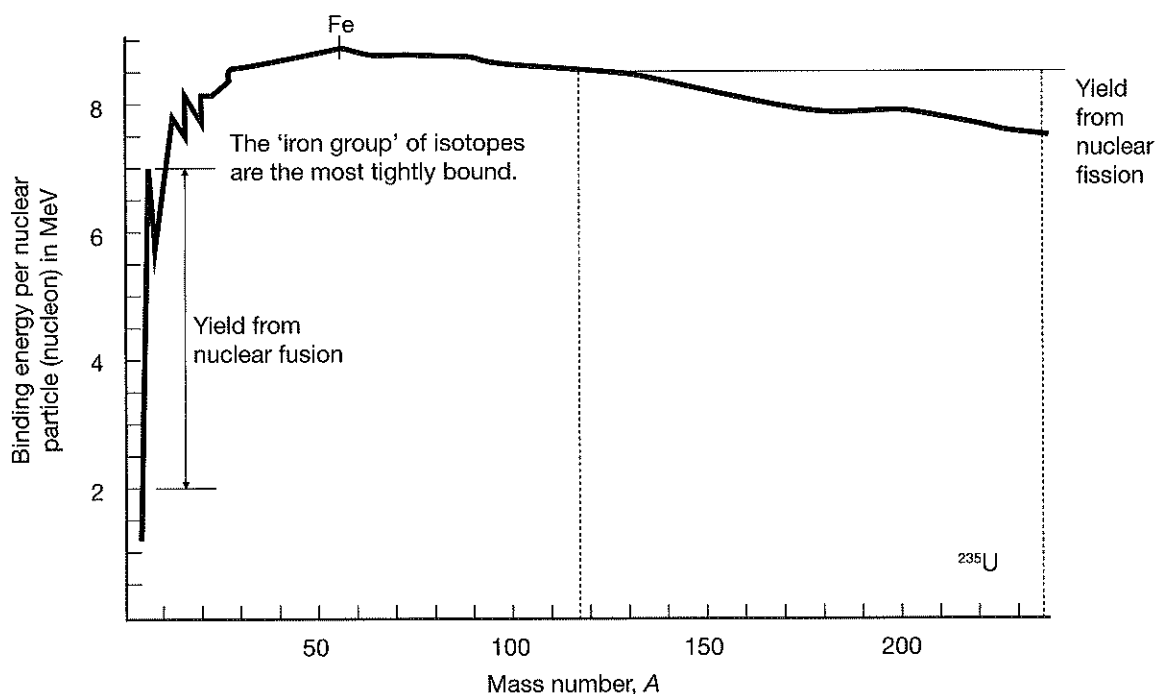
(a) Without making any calculations, which nucleus would have the greater total binding energy? Justify your answer.

(b) Which has the greater binding energy per nucleon?

(c) On this information alone, which of these two nuclei would be the most stable? Justify your answer.



2.7.1.4 Consider the following graph.



(a) According to this graph, which is the most stable nucleus? Justify your answer.

(b) This graph suggests that elements with a mass number less than 56, the mass number of iron, will, if they undergo a nuclear reaction, undergo nuclear fusion and not nuclear fission. Explain this prediction.

(c) This graph suggests that elements with a mass number greater than 56, the mass number of iron, will, if they undergo a nuclear reaction, undergo nuclear fission and not nuclear fusion. Explain this prediction.

2.7.2 Mass defect and nuclear energy.

2.7.2.1 Define mass defect.

2.7.2.2 Explain the connection between binding energy and mass defect.

2.7.2.3 A helium nucleus contains two protons and two neutrons. Which of the following statements is true?

- (A) The mass of the nucleus is equal to the mass of the protons plus the mass of the neutrons.
- (B) The mass of the nucleus is more than the mass of the protons plus the mass of the neutrons.
- (C) The mass of the nucleus is less than the mass of the protons plus the mass of the neutrons.
- (D) The mass of the nucleus is equal to the mass of the protons plus the mass of the neutrons plus the binding energy.

2.7.2.4 What is meant by the 'equivalence of mass and energy'?

- (A) Mass and energy can be considered to be different forms of matter and are interchangeable.
- (B) The laws of conservation of both mass and energy hold in any chemical or nuclear process.
- (C) Mass is the same as energy.
- (D) Mass can be converted into energy in a nuclear process.

2.7.2.5 The mass defect of a nucleus is 0.004356 u. What is its energy equivalent?

- (A) 6.5×10^{-13} MeV
- (B) 4.06 MeV
- (C) 4.06 J
- (D) 6.5×10^{-13} J

2.7.2.6 What, in joules, is the energy equivalent of 0.5 grams of iron?

- (A) 4.5×10^6 J
- (B) 4.5×10^7 J
- (C) 4.6×10^{14} J
- (D) 4.5×10^{17} J

2.7.2.7 A nuclear transmutation released 168 MeV of energy. What is the mass defect for this reaction?

- (A) 0.018035 u
- (B) 0.001803 g
- (C) 2.688×10^{-17} u
- (D) 0.18035 u

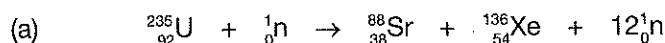
2.7.2.8 A nuclear transmutation released 5.6×10^{-14} J of energy. What is the mass defect for this reaction?

- (A) 3.746×10^{-4} u
- (B) 5.2165×10^{-11} u
- (C) 5.83×10^{-28} u
- (D) 5.994×10^{-31} u

2.7.2.9 Use the information about isotopes of some elements in the data table to answer this question.

Particle	Mass (amu)	Particle	Mass (amu)
$^{12}_6\text{C}$	12.00000	^7_3Li	7.0160
^1_1H	1.007825	^1_0n	1.0087
^2_1H	2.014102	$^{15}_7\text{N}$	15.000109
^4_2He	4.0026	$^{88}_{38}\text{Sr}$	87.9056
$^{92}_{36}\text{Kr}$	91.8973	$^{235}_{92}\text{U}$	235.0439
		$^{136}_{54}\text{Xe}$	135.9072

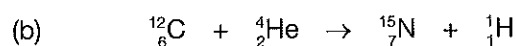
Find the mass defect in each of the following reactions.



.....

.....

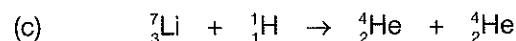
.....



.....

.....

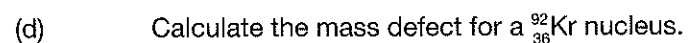
.....



.....

.....

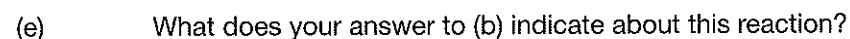
.....



.....

.....

.....



.....

.....

.....

2.8 Alpha and beta decay, spontaneous and artificial transmutations.

2.8.1 Spontaneous and artificial transformations.

2.8.1.1

(a) Clarify the idea of a spontaneous nuclear reaction.

(b) Which radioisotopes undergo spontaneous transformation? Explain why.

2.8.1.2

(a) Clarify the idea of an artificial nuclear reaction.

(b) Why are most artificial transformations done?

(c) Who carried out the first artificial transformation and when?

(d) What important discovery did this experiment (answer (c)) eventually lead to?

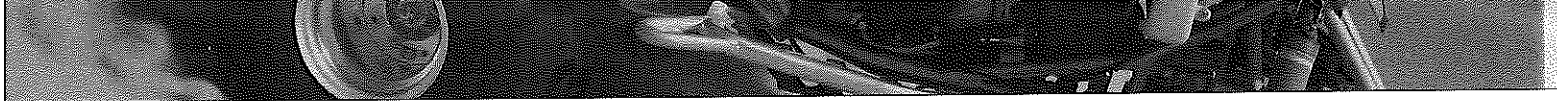
2.8.1.3

(a) The first particles used to bombard atomic nuclei to study atomic structure and to induce artificial nuclear reactions were positively charged protons and alpha particles. Explain why.

(b) Identify two limitations of using these particles to bombard nuclei.

(c) Explain why most artificial transformations done in laboratories in more modern times use neutrons as the 'bullet' to split the target nucleus apart.

(d) What is the major limitation of using neutrons as bullets in nuclear transformations?



2.8.1.4 In 1930 Bothe and Becker used polonium-210 as an alpha particle source to bombard thin sheets of boron, lithium and beryllium metals in their experiments to study radioactivity. The diagram simulates their experiment.

Using your knowledge to determine the nuclide produced in each transformation, write an equation for the experiment if the metal used was:

(a) Aluminium.

.....

.....

(b) Magnesium.

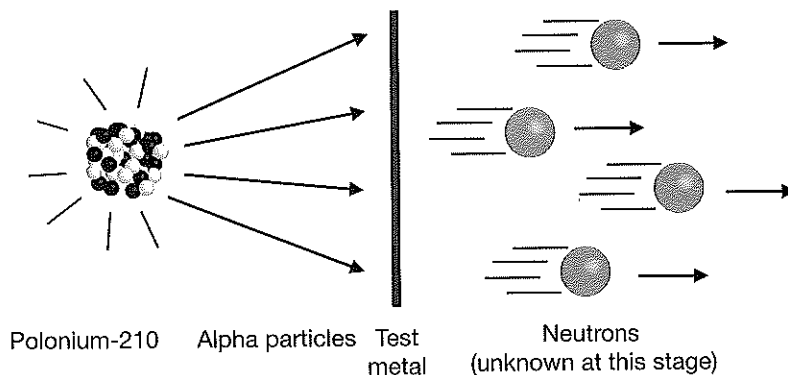
.....

.....

(c) Sodium.

.....

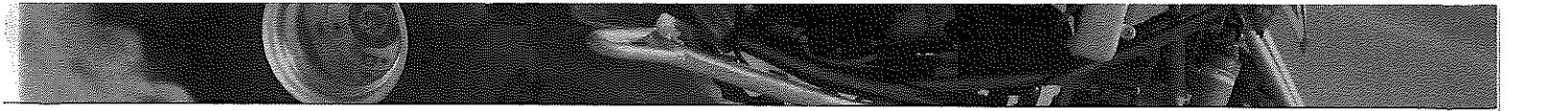
.....



Walter Bothe.



Herbert Becker.



2.9 Neutron induced nuclear fission.

2.9.1 Neutron induced fission.

2.9.1.1 What is involved in neutron induced fission?

.....

.....

.....

2.9.1.2 Why was the discovery of the neutron significant for artificial nuclear transformations?

.....

.....

.....

2.9.1.3 One observation of neutron induced nuclear transformations puzzled scientists. What was this observation?

.....

.....

2.9.1.4 Outline the two different types of transformation different speed neutrons were observed to cause.

.....

.....

.....

.....

.....

.....

2.9.1.5 What did this observation and prediction lead to?

.....

.....

.....

.....

2.9.1.6 By 1939 it was known that during neutron induced transformations, a small number of neutrons were often released as a product. Why did this excite scientists at the time?

.....

.....

2.9.1.7

- (a) When a neutron is absorbed into a nucleus, that nucleus does not necessarily become a new element. It may become a heavier isotope of that element. Clarify this statement using as your example an atom of N-14 being bombarded with a neutron. Include the equation for the transformation.

.....

.....

.....

- (b) In gaining an extra neutron an element might become a radioactive isotope. It would then break down by emitting a beta particle. Clarify this by considering K-35 gaining a neutron to form X which then undergoes beta decay, to form Y. Identify X and Y. Include equations for the reactions.

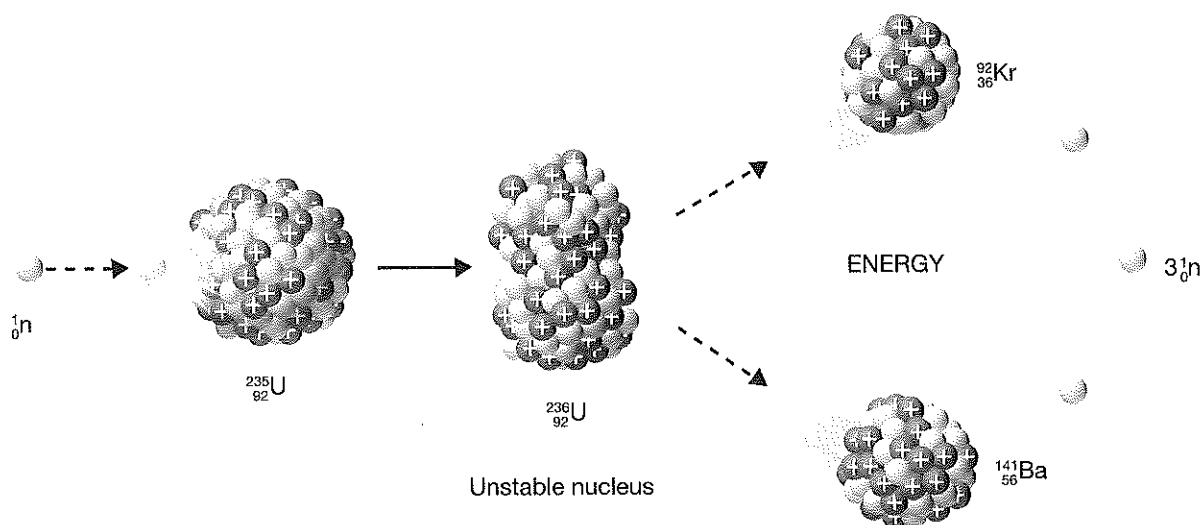
.....

.....

.....

.....

2.9.1.8 The diagram shows the neutron induced fission of U-235.



- (a) Write the nuclear equation for the neutron capture reaction.
-
- (b) Write the equation for the fission reaction for the product of the neutron capture.
-
- (c) What is the significance of the additional neutrons released in the fission reaction?
-

2.10 Fission chain reactions.

2.10.1 Controlled and uncontrolled chain reactions.

2.10.1.1

(a) Distinguish between a controlled and an uncontrolled nuclear reaction.

(b) How does this difference affect the energy released in the processes?

(c) Assuming the same nuclear fission reaction occurs for both a controlled and an uncontrolled nuclear reaction, what happens in order to control the reaction?

2.10.1.2

(a) On average, as determined by Fermi, how many neutrons per fission are required to sustain a controlled nuclear reaction?

(b) Why this number? Surely only one neutron could initiate the next fission.

(c) What would happen to the nuclear reaction if, on average, only 0.75 neutrons per fission had sufficient energy to initiate further fission reactions?

2.10.1.3

(a) Write nuclear equations for each of the following reactions (note that you will need to include emitted neutrons and beta particles if appropriate).

(i) Neutron capture by U-235, then the product's fission to produce Ba-141 and Kr-92.

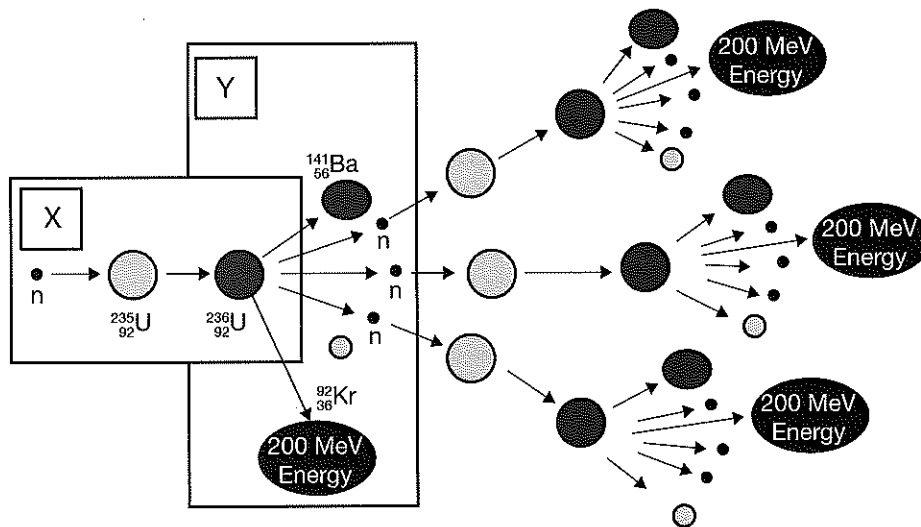
(ii) Neutron capture by U-235, then the product's fission to produce La-139 and Mo-95.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 104

.....

.....

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 104

[illegible]

.....

.....

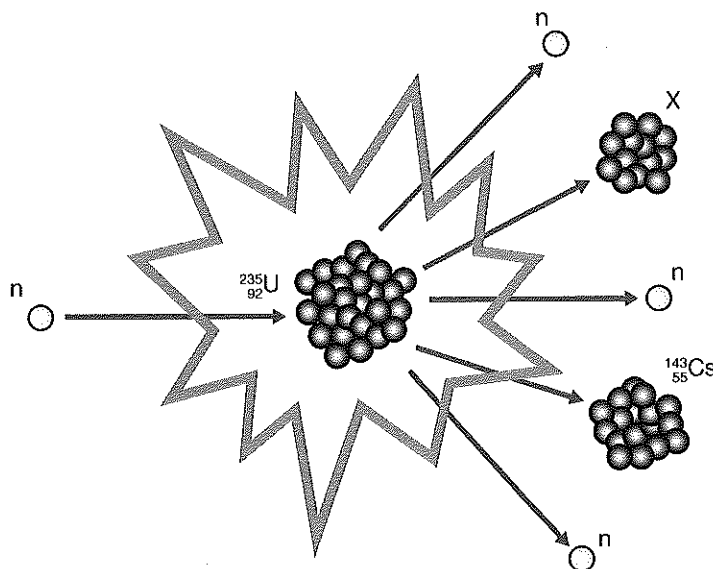
2.10.2 Nuclear fission reactions.

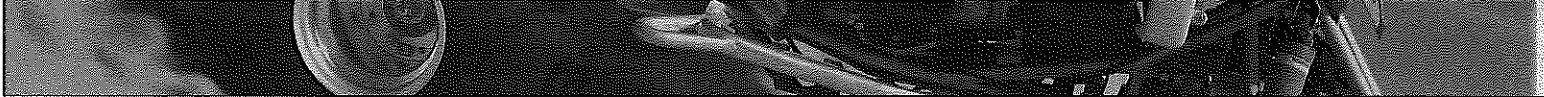
2.10.2.1

- (a) Clarify the concept of nuclear fission.
- (b) What is the source of the energy released during nuclear fission reactions?
- (c) In general, which elements undergo nuclear fission?
- (d) Explain why it is these elements that undergo fission rather than fusion.

2.10.2.2 The diagram represents the fission reaction of U-235 following neutron bombardment.

- (a) Identify the unlabelled nuclide X.
- (b) Write the nuclear equation for the fission reaction (be careful).
- (c) One product has been omitted from the diagram. What is it?
- (d) What is the significance in this particular reaction of the release of three neutrons as product?





2.11 Nuclear fusion.

2.11.1 Nuclear fusion.

2.11.1.1

- (a) Clarify the concept of nuclear fusion.
.....
- (b) What is the source of the energy released during nuclear fusion reactions?
.....
- (c) In general, which elements undergo nuclear fusion?
.....
- (d) Explain why it is these elements that undergo fusion rather than fission.
.....
.....
.....
.....
.....
.....

2.11.1.2 State the conditions required for a spontaneous nuclear fusion reaction to occur.

.....
.....
.....

2.11.1.3 Consider these two statements made by students to define binding energy.

Statement 1: Binding energy is the energy required to split an atom into its component parts.

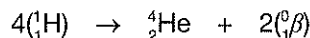
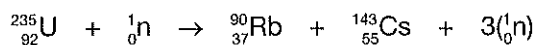
Statement 2: Binding energy is the energy released when the nucleons that make up a nuclide combine to form that nuclide.

Which student is correct? Justify your answer.

.....
.....
.....
.....

2.11.2 Energy in fission and fusion.

2.11.2.1 The equations show a U-235 fission reaction and a simplified version of the hydrogen fusion reaction occurring in the Sun. The table shows the masses of the nucleons and isotopes involved in these reactions.



(Note that the hydrogen fusion in the Sun occurs in several steps, the equation above simply summarises the net result in one.)

Particle	Mass (amu)
${}_1^1\text{H}$	1.007276
${}_0^1\text{n}$	1.008664
${}_2^4\text{He}$	4.002602
${}_{92}^{235}\text{U}$	235.04393
${}_{37}^{90}\text{Rb}$	89.91380
${}_{55}^{143}\text{Cs}$	142 92735
${}_1^0\beta$	0.0005486

(a) Complete the following table by calculating the quantities indicated.

(a)	Mass defect for the U-235 fission reaction in amu	
(b)	Energy released in MeV	
(c)	Mass defect for the U-235 fission reaction in kg	
(d)	Energy released in joules	
(e)	Mass of one U-235 atom in kg	
(f)	Energy released per kg of U-235 fission	
(g)	Mass defect for the hydrogen fusion reaction in amu	
(h)	Energy released in MeV	
(i)	Mass defect for the hydrogen fusion reaction in kg	
(j)	Energy released in joules	
(k)	Mass of fusion reactants in kg	
(l)	Energy released per kg of fusion atoms	

(b) Comment on the relative amounts of energy released in the fusion and fission reactions.

(c) Explain why the energy amounts calculated in (b) and (h) or (d) and (j) were not used to compare these processes.

2.12 The use of scientific knowledge and international collaboration.

2.12.1 The Manhattan project.

Before and during the two world wars there was a great deal of competition between scientists in Europe and between physicists in England and Germany in particular, to the point that new discoveries were often not accepted by the rival groups.

In addition, especially as World War II approached, the application of scientific study to 'the war effort' was a problem for many scientists.

Two groups emerged – those who were appalled at the concept of turning their ideas into weapons of destruction, and those who thought scientific study was justified in its own right, and decisions of what to do with the results should be left to the politicians.

Planck and Einstein were both German physicists. They devoted their lives to science and were innovative and intellectually creative. Both had enormous influence on science. Both were at the forefront of the start of the change from classical to quantum physics. Both were working in Germany at the start of World War I.

Planck was a nationalist; he believed in his country and worked for his country. He was loyal to whatever the government wanted. When war broke out, Planck was one of the first of ninety-three German intellectuals to sign a document supporting the role of Germany in the war.

Along with the others, he then devoted his work and research to whatever the war effort required of him. Einstein was different. He had no particular politics and supported no government. He had moved out of Germany at a young age and most of his work was based in Switzerland. He became a Swiss citizen.

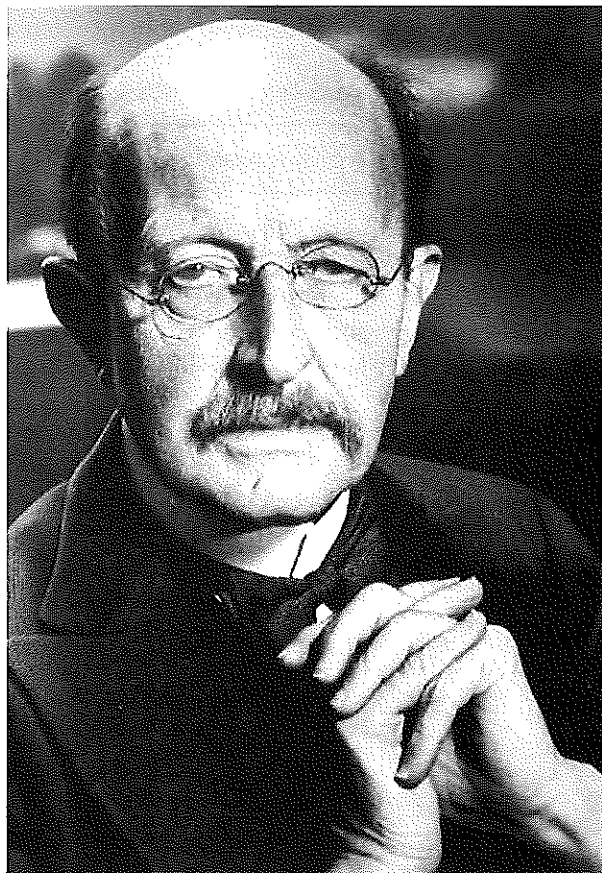
Although of German descent, and working in Germany at the time, Einstein refused to sign the document supporting Germany's war effort, instead signing a different document which called for a peaceful world.

He was a pacifist, and did not accept that his work should contribute to the killing of others. This aspect of the science/society debate was emphasised after Adolf Hitler's rise to power in Nazi Germany, mainly due to his ideology of Aryan purity.

This resulted in all scientists of non-German descent being dismissed and their work classed as not 'true science'. In 1933, along with the works of many famous people, Einstein's books were burnt in the streets of Berlin.

Einstein's fears grew in the early 1940s when he realised that his work (he had predicted that an enormous amount of energy was available in matter if it could be accessed) was being researched further in Germany. He wrote to the then president of the United States urging similar developments occur in America – and so the Manhattan project to develop a nuclear bomb was born. Had this not occurred, the results of World War II may have been quite different.

Even then there were scientists who refused to be involved, and those who rushed to be part of it.



Max Planck.

2.12.1.1 Discuss the impact of politics on each of Einstein's and Planck's work.

.....

.....

.....

.....

.....

.....

2.12.1.2 What might have been the attitudes of ordinary people in German society towards the attitudes of Einstein and Planck in this situation?

.....

.....

.....

2.12.1.3 Identify the major technology that developed because of the Manhattan project and assess its impact on our society for the following periods.

(a) During the 1940s.

.....

.....

.....

.....

(b) Over the subsequent 30 years.

.....

.....

.....

.....

(c) At the present time.

.....

.....

.....

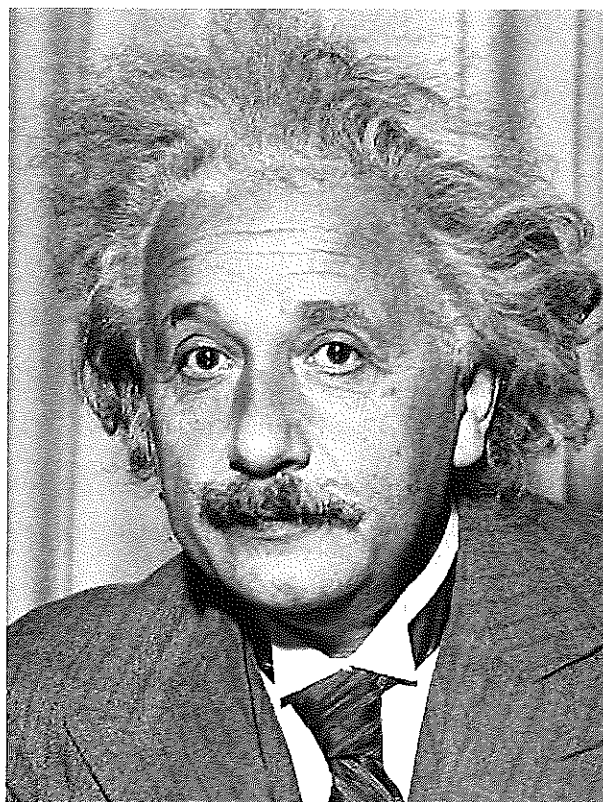
.....

2.12.1.4 Einstein wrote, 'I made one great mistake in my life – when I signed that letter to President Roosevelt recommending that atom bombs can be made.' Suggest why he might have written these words.

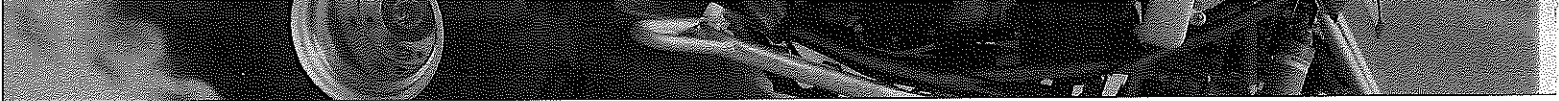
.....

.....

.....



Albert Einstein.



2.12.2 Nuclear waste.

2.12.2.1

- (a) It is actually quite difficult to get an accurate measure of the amount of nuclear waste produced in modern times. Give reasons for this difficulty.

.....

.....

.....

- (b) On what two properties is radioactive waste classified?

.....

- (c) Would it be reasonable to classify radioactive waste on just one of these properties? Justify your answer.

.....

.....

.....

2.12.2.2 Give a brief description of each of the following types of radioactive wastes, and the relative radioactivity of each type. You may research the information you need to complete this work.

- (a) Uranium tailings.

.....

.....

- (b) Exempt waste and very short lived waste.

.....

.....

.....

- (c) Low level waste.

.....

.....

.....

.....

.....

- (d) Intermediate level waste.

.....

.....

.....

.....

.....

(e) High level waste.

(f) Transuranic waste.

2.12.3 The Australian nuclear waste situation.

2.12.3.1 Research the situation in Australia regarding the disposal of the small amounts of waste materials produced at the Lucas Heights reactor in Sydney, New South Wales during its day to day operation and its production of radioisotopes for medical, industrial and agricultural use. Be sure to include the relevant government regulations for the disposal of this waste.

Word process your report and summarise your main points here.

2.13 Development of models and theories.

2.13.1 The birth of stars.

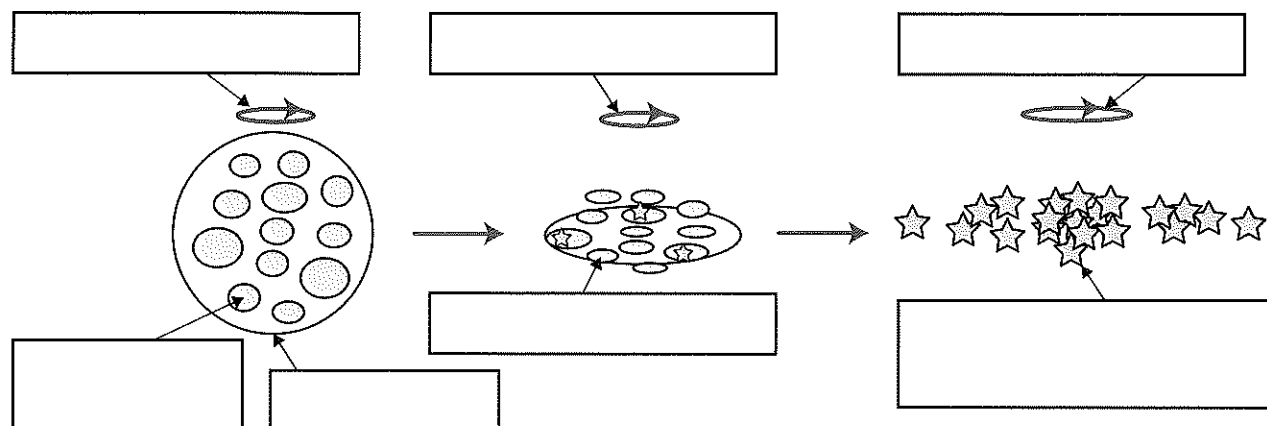
2.13.1.1 What is the currently most accepted theory for the beginning of the Universe and the subsequent birth of stars and galaxies?

2.13.1.2 The sentences in the table summarise the sequence of events thought to have occurred following the event identified in Question 2.13.1.1. However, only the first half of each sentence is in order, the second halves being mixed up. Match each second half to its correct first half to complete each sentence.

	First half – in order	Second half – not in order
(a)	The Big Bang produced both matter	left a 'lumpy' Universe.
(b)	The annihilation of some matter/antimatter	matter in the 'lumps' together.
(c)	This consisted of pockets of unevenly distributed matter with enormous	at the more dense parts.
(d)	Gravitational forces drew the	nuclear fusion reactions.
(e)	Individual matter clouds which we call	matter started to form within the larger matter cloud.
(f)	The denser parts exerted greater gravitational	and antimatter with matter in excess.
(g)	More matter began to accumulate	cloud, its speed of rotation increased.
(h)	These clouds of matter were thought to be rotating	becoming compressed and hotter.
(i)	Small, but growing clumps of denser	empty spaces where annihilation had occurred.
(j)	As more matter started to be drawn closer into the	and galaxies formed.
(k)	This spinning motion caused the cloud	protogalaxies, which were also 'lumpy', formed.
(l)	As their size and density increased the 'lumps' of matter collapsed further,	slowly as a result of the forces associated with the Big Bang event.
(m)	Eventually a core was hot enough to sustain	to flatten out and form a disc.
(n)	Stars, composed of hydrogen were born,	attraction on the less dense parts.

2.13.1.3 The diagram below represents part of the process described by the sentences in the table. It is missing labels. Complete the diagram by placing these labels in appropriate positions.

Cloud flattens due to spin, Fusion forms stars within galaxy, Lumps within the cloud, Matter cloud, Rotation of galaxy, Rotation of matter cloud, Rotation of protogalaxy



2.13.1.4 Why is the Big Bang theory for the development of the Universe accepted over other proposed theories?

2.13.2 The Hertzsprung-Russell diagram.

2.13.2.1 What is an HR diagram?

2.13.2.2 What new information did the first HR diagram show astronomers?

2.13.2.3 The x-axis on HR diagrams can show, either individually, or at the same time, the colours, spectral classes and temperatures of stars. Explain why.

2.13.2.4 Compare the luminosity of a star with its brightness.

2.13.2.5

(a) Which group of stars is very hot but has low luminosity?

(b) What is the source of energy of these stars?

2.13.2.6

(a) Which group of stars has high luminosity but low temperature?

(b) Explain how this seeming contradiction is actually true.



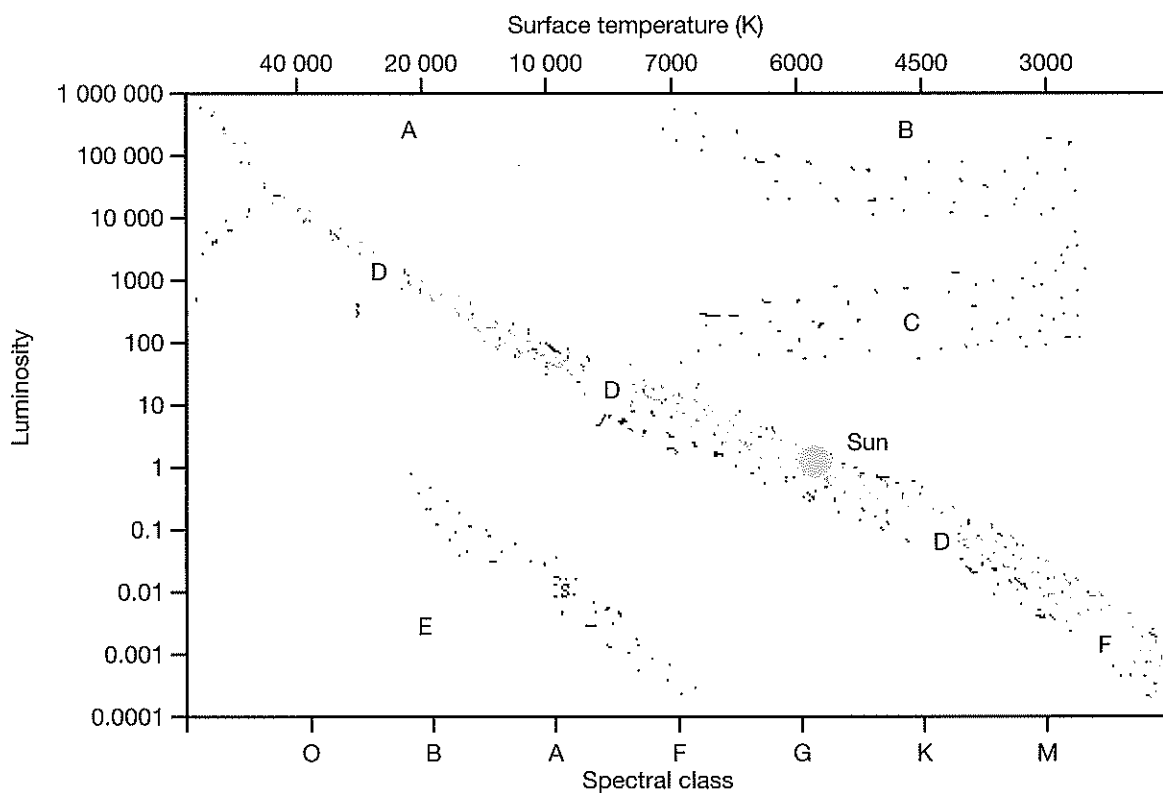
2.13.2.7 What does knowing the position of a star on the Hertzsprung-Russell diagram tell us about the star?

.....

.....

2.13.2.8 Identify the labels required for the diagram below.

- A
- B
- C
- D
- E
- F



2.13.2.9 Why is measuring the luminosity of stars difficult?

.....

2.13.2.10

(a) If a star has a luminosity of ten thousand (10 000), how many times brighter is it than the Sun?

.....

(b) How does the brightness of white dwarfs relate to that of the Sun?

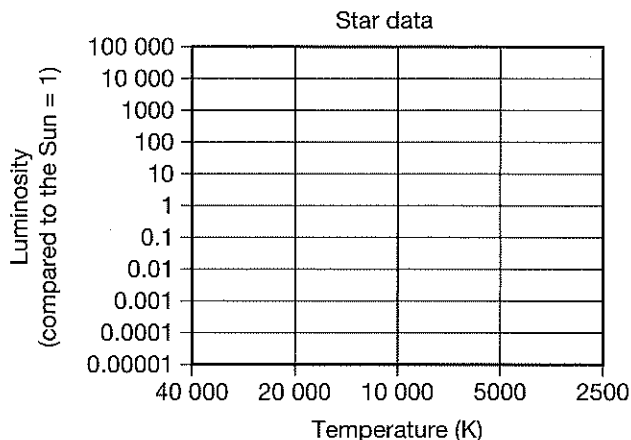
.....

(c) What type of star would have a higher luminosity than the Sun, but a lower temperature?

.....

2.13.3 The Hertzsprung-Russell diagram – a practical.

2.13.3.1 Using either the graph grid provided, or a spreadsheet program, plot the star data from the table onto a Hertzsprung-Russell diagram.



	Star	Surface Temperature (K)	Luminosity compared to the Sun
A	The Sun	5860	1
B	Proxima Centauri	3240	0.00006
C	Alpha Centauri A	5860	1.6
D	Alpha Centauri B	5250	0.45
E	Sirius A	9230	23.5
F	Sirius B	9000	0.003
G	Canopus	7700	1400
H	Betelgeuse	3450	12 600
I	Rigel	11 200	42 000
J	Beta Centauri	24 000	3500
K	Wolf 349	2640	0.00002
L	Barnard's Star	3240	0.00045
M	Arcturus	4420	110
N	Procyon	6440	7.2
O	Archenar	15 400	200
P	Alderbaran	15 400	95

2.13.3.2 In the table provided classify each star as main sequence, red giant, white dwarf or supergiant.

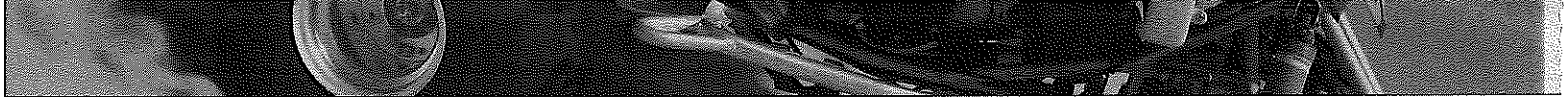
2.13.3.3 Identify the most luminous star.

2.13.3.4 Identify the star with the lowest surface temperature.

2.13.3.5 Identify the star with the highest surface temperature.

2.13.3.6 Explain why this star doesn't have the highest luminosity.

Classification of stars		
A	The Sun	
B	Proxima Centauri	
C	Alpha Centauri A	
D	Alpha Centauri B	
E	Sirius A	
F	Sirius B	
G	Canopus	
H	Betelgeuse	
I	Rigel	
J	Beta Centauri	
K	Wolf 349	
L	Barnard's Star	
M	Arcturus	
N	Procyon	
O	Archenar	
P	Alderbaran	



2.13.4 Main sequence stars.

2.13.4.1

- (a) What is the distinguishing characteristic of main sequence stars?
-
- (b) What position do main sequence stars occupy on the Hertzsprung-Russell diagram?
-
- (c) Where are the coolest and the hottest main sequence stars found on the Hertzsprung-Russell diagram?
-
- (d) Where are the most and least luminous main sequence stars found on the Hertzsprung-Russell diagram?
-
- (e) Which stars on the main sequence will stay there the shortest time before going to the next stage in their evolutionary path? Explain why.
-

2.13.4.2 Which of the following sequences of spectral types represent the coolest to hottest stars?

- (A) MKGFABO
- (B) ABFGKMO
- (D) OBAFGKM
- (D) MFKGABO

2.13.4.3 Which of the following main sequence stars will be both the hottest and the largest?

- (A) Red dwarf
- (B) White dwarf
- (C) Red giant
- (D) Blue giant

2.13.4.4 What proportion of our Sun undergoes hydrogen fusion to produce the energy it emits?

- (A) About 5%
- (B) About 10%
- (C) About 25%
- (D) About 40%

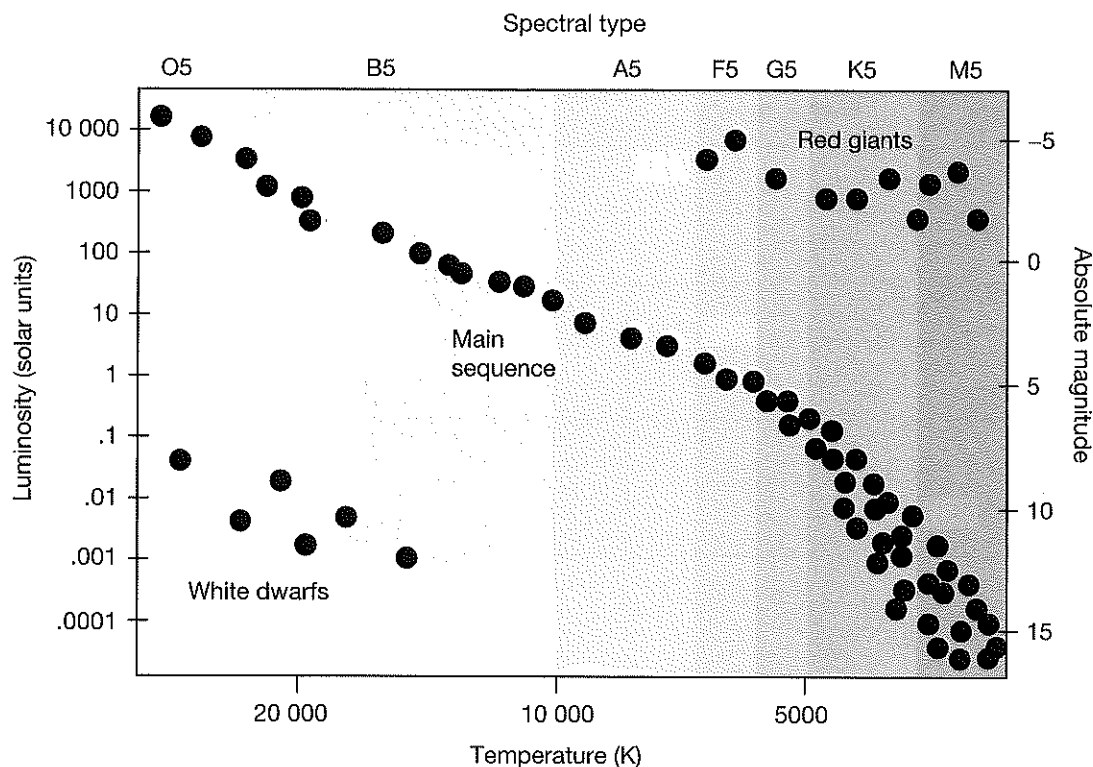
2.13.4.5 Star X is a main sequence star that appears red when viewed from Earth. Which of the following statements about X is true?

- (A) It will be hotter than our Sun.
- (B) It will appear brighter than a white dwarf.
- (C) It will live longer than our Sun.
- (D) It will be the same size as a red giant star of the same temperature.

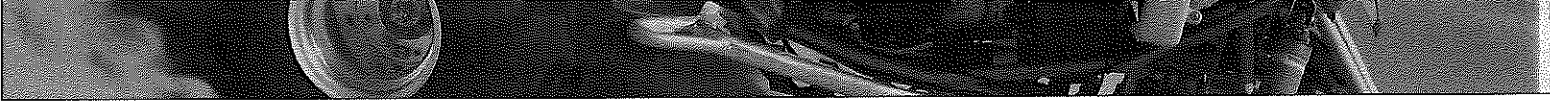
2.13.4.6 In a main sequence star, gravitational collapse is balanced by:

- (A) Convection of stellar material from the core.
- (B) Outwards pressure caused by energy produced during nuclear fusion.
- (C) Outwards pressure from coronal mass ejections from the core.
- (D) Interior cooling of the star.

Use this Hertzsprung diagram to answer the next FOUR questions.



- 2.13.4.7** Which is hotter, a main sequence star with an absolute magnitude of $M = 8$ or a white dwarf with a luminosity 100 times smaller than the Sun?
- The main sequence star.
 - The white dwarf.
 - They have the same temperature.
 - There is insufficient information to determine this.
- 2.13.4.8** Which statement is the most correct about the comparison between a spectral type G5 main sequence star and a spectral type O5 main sequence star?
- The G5 star is cooler, dimmer, smaller, and will not live as long as the O5 main sequence star.
 - The G5 star is hotter, brighter, smaller, and will live longer than the O5 main sequence star.
 - The G5 star is smaller, hotter, brighter, and will not live as long as the O5 main sequence star.
 - The G5 star is dimmer, cooler, smaller, and will live longer than the O5 main sequence star.
- 2.13.4.9** Consider the information given below about three main sequence stars A, B, and C.
- Star A has an absolute magnitude of 0.
 - Star B has the same luminosity as the Sun.
 - Star C has a spectral type of M5.
- Which of the following is a true statement about these stars?
- Star A has the greatest mass.
 - Star B has the greatest mass.
 - Star C has the greatest mass.
 - Stars A, B and C all have approximately the same mass.
- 2.13.4.10** At which grid point (spectral class and absolute magnitude) on this diagram would we find our Sun?
- K5, 0.1
 - G5, 1
 - G5, 5
 - F5, 5



2.13.5 Non main sequence stars.

2.13.5.1 Complete the table to summarise information about non main sequence stars.

Star type	Energy source	Where found on HR diagram	Main features

2.13.5.2

- (a) Explain why neutron stars and black holes do not appear on an HR diagram.
.....
.....
- (b) Some resources write about blue giants and blue supergiants while others, like this book only write about blue supergiants. Why the difference?
.....
.....
.....
- (c) Explain why there are two groupings of red giants – the giants and the supergiants.
.....
.....

2.13.5.3 Explain why shell burning in red giants and red supergiants (and blue supergiants) is able to produce elements up to and including iron, but no elements past iron on the periodic table.

2.13.5.4 Elements beyond iron do occur in some stars, like our Sun. Explain how these are formed.

2.13.5.5 Our Sun is at least a 'second generation' star.

(a) What does this mean?

(b) What evidence do astronomers base this classification on? Explain their reasoning.

2.13.5.6

(a) What is a black dwarf star?

(b) Why are there thought to be no black dwarfs in the Universe?

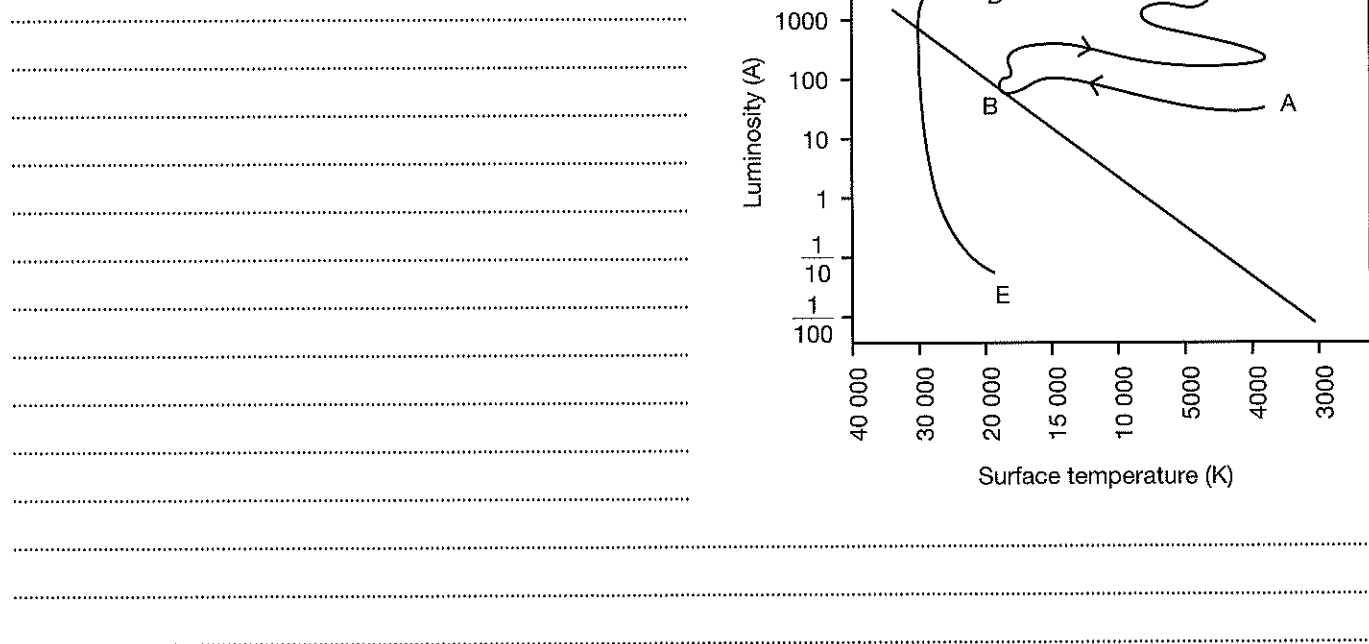
(c) If black dwarf stars have not been discovered, why are they thought to be a possibility?

(d) If there were black dwarf stars, why would they be extremely difficult to discover?

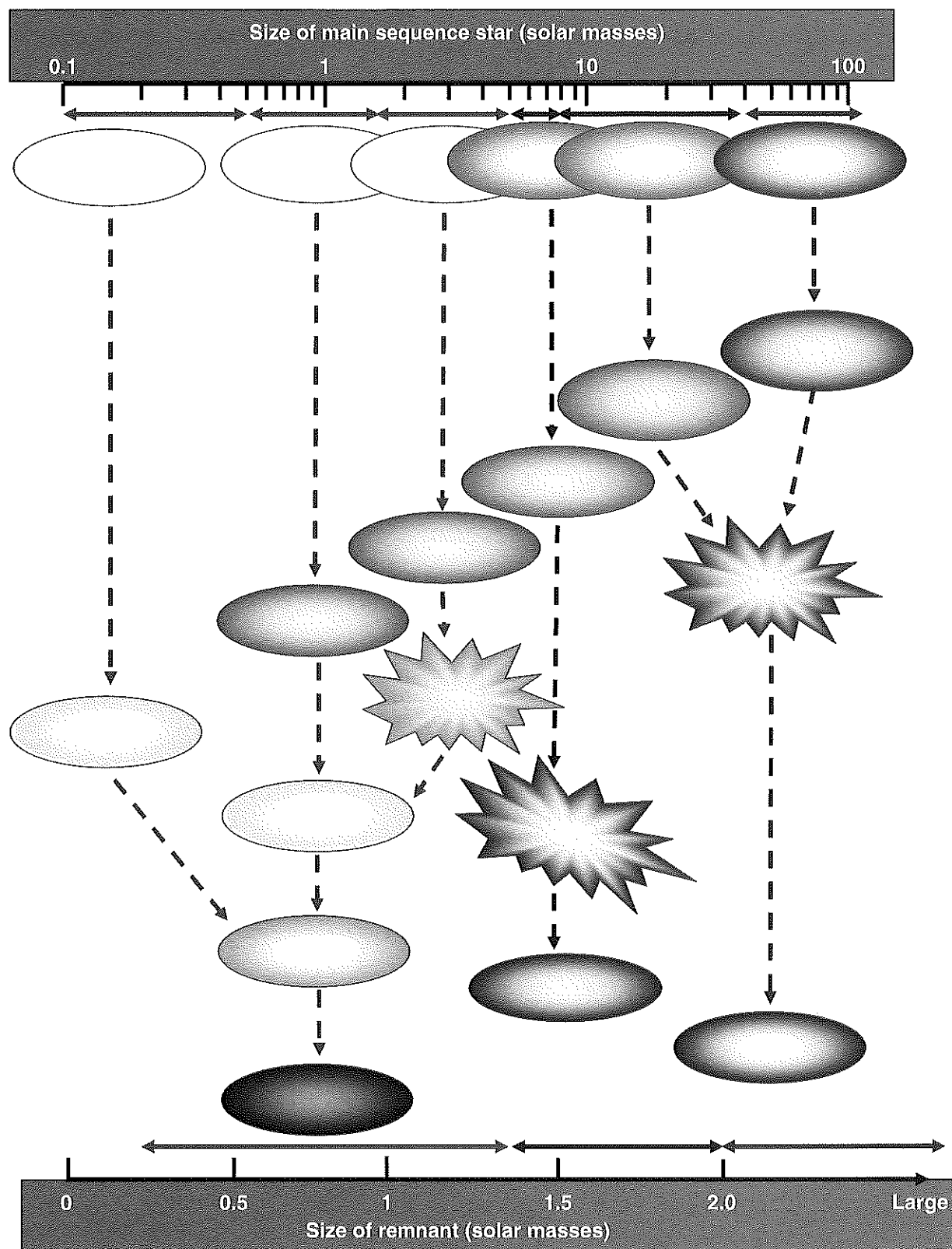
2.13.5.7 If black holes do not emit any radiation then how are they detected?

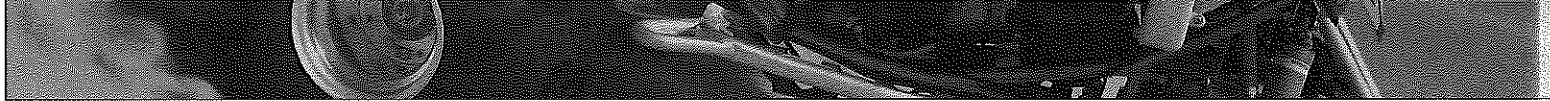
	★ Star P		
	★ Star Q		
	Star R ★		
		Star S ★	
			Star T ★

- 2.13.6.2** The diagram shows the evolutionary pathway of a star. Summarise this pathway identifying the type of star at each labelled position and the energy source of the star at each position.



2.13.6.3 The diagram summarises the evolution of main sequence stars, but all the labels are missing. Write in the labels either on this diagram or a copy you make.





2.13.6.4 Outline the evolutionary pathway of a star which has a mass between 0.5 and 1.4 solar masses.

.....

.....

.....

.....

.....

2.13.6.5 Outline the evolutionary pathway of a star which has a mass greater than 40 solar masses.

.....

.....

.....

.....

.....

.....

2.13.6.6 How do we know that our Sun is a second (at least) generation star?

- (A) It is in the middle of the main sequence.
- (B) It is relatively small compared to most main sequence stars.
- (C) Its core is not yet hot enough to fuse helium.
- (D) It contains elements with higher atomic mass than iron.

2.13.6.7 What is the main reason that red giant stars fuse elements larger than helium in layers surrounding their cores, but main sequence stars do not?

- (A) They are larger.
- (B) They have more mass.
- (C) They are hotter.
- (D) They are more stable.

2.13.6.8 Which of the following lists, in correct order, a possible evolutionary path for a star with a mass 0.3 times the mass of our Sun?

- (A) Main sequence star, white dwarf.
- (B) Main sequence star, red giant, white dwarf.
- (C) Main sequence star, red giant, nova, black hole.
- (D) Main sequence star, red giant, supernova, neutron star.

2.13.6.9 What will be the most likely final stage in the evolutionary path of a star with three times the mass of the Sun?

- (A) A black hole.
- (B) A neutron star.
- (C) A supernova explosion.
- (D) A white dwarf.